

PONIR KUMER SAHA

Principal Engineer | System Architect | Senior Java | Spring Boot Backend Engineer | Microservices | Kafka | AI/RAG
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Dhaka, Bangladesh

Summary

Results-driven Principal Engineer and System Architect with 13+ years of experience leading engineering teams and delivering cloud-native, distributed, AI-powered, and high-availability platforms.

Proven expertise in scalable backend systems, microservices, event-driven architecture, Retrieval-Augmented Generation (RAG), LLM integration, vector search, real-time processing, cloud infrastructure, and AI-assisted application development.

Experienced in defining technical strategy, designing enterprise AI solutions, mentoring engineers, driving engineering excellence, and partnering with stakeholders to deliver business-critical products. Strong background across financial services, education technology, navigation, logistics, and enterprise platforms.

Core Competencies

Architecture & Design:	Distributed Systems, Event-Driven Architecture, Kafka, CQRS, Outbox Pattern, Debezium CDC, DDD, SOA
Cloud & DevOps:	AWS, GCP, Docker, Kubernetes, Jenkins, CI/CD, Prometheus, Grafana, OpenTelemetry, Distributed Tracing
Backend Engineering:	Java, Spring Boot, Spring Cloud, Spring Security, TypeScript
Databases & Search:	PostgreSQL, MySQL, Oracle, MongoDB, Redis, OpenSearch, Vector Search
AI & LLM Engineering:	Spring AI, LangChain4j, RAG, LLM Integration, Prompt Engineering, AI Agents, Langfuse, OpenAI, Gemini
Leadership & Delivery:	Technical Architecture, Code Review, Mentoring, Stakeholder Communication, Agile/Scrum, Delivery Ownership

Professional Experience

Allianz Technology	Remote
Senior Full Stack Developer	04/2026 - Current
- Designed and implemented AI-assisted case management modernization for an insurance platform, migrating legacy workflows into a scalable Spring Boot microservices architecture. - Integrated AI/RAG capabilities using OpenSearch, vector embeddings, OpenAI/Gemini, and document intelligence to support contextual case search, policy document retrieval, and intelligent assistant workflows. - Improved case handling efficiency by reducing manual document review, accelerating response time, and enabling more consistent knowledge access for business users.	
BJIT Limited	Dhaka, Bangladesh
Principal Software Engineer / Architect	02/2022 - 03/2026
- Architected microservices-based platforms with Spring Cloud, reducing API latency by 40% and improving scalability. - Designed real-time event-driven architectures using Kafka, Redis, and WebSocket, delivering low-latency communication for high-volume distributed systems. - Led cross-functional engineering teams of 10+ developers, architects, QA, and DevOps engineers, driving architecture decisions, technical governance, delivery planning, and mentoring engineers.	
Dnet - A Social Enterprise	Dhaka, Bangladesh
Lead Java Developer	11/2020 - 02/2022
- Led the architecture and implementation of event-driven architecture with Kafka, enhancing throughput and overall system reliability. - Directed cloud migration strategy, achieving 99.9% uptime and reducing infrastructure costs. - Collaborated with leadership to align technical roadmaps with organisational objectives.	

- Developed and maintained backend APIs using Java Spring Boot and Node.js for cloud-native applications.
- Standardised testing frameworks, raising code coverage to **80%**.

- Designed **secure financial systems**, reducing downtime by 20% and enhancing transaction reliability.
- Refactored core modules, improving **DB query performance by 35%**.
- Ensured compliance with **financial data security standards**

- Automated data transformation processes, ensuring 99.9% accuracy in logistics operations.
- Engineered custom PDF generation tools to automate and streamline financial document workflows, cutting manual processing time significantly.

Education

Certifications

- Oracle Certified | Java SE 7 Programmer
- Certified ScrumMaster® | Scrum Alliance

Highlighted Projects

1. Learning Orbit – AI-Powered Learning Management Platform | SaaS

Tech Stack: Spring Boot, Angular, PostgreSQL, OpenSearch, Spring AI, LangChain4j, Gemini, OpenAI, OCR, WebSocket

GitHub: <https://github.com/ponir-saha/learning-orbit>

 - Architected and developed an AI learning platform that converts approved PDF textbooks into chapter-aware study, AI tutoring, practice exams, and teacher-reviewed assessments.
 - Built RAG-based learning workflows with contextual tutoring, OCR support, chapter extraction, intelligent assessment generation, and instant feedback
 - Implemented secure student, teacher, parent, and admin workflows with real-time progress tracking and data-driven learning insights.
2. Itinerary Builder - Travel SaaS Operations Platform | SaaS

Tech Stack: Spring Boot, Angular, PostgreSQL, Spring Security JWT, Flyway, Docker Compose, PDF Generation

GitHub: <https://github.com/ponir-saha/itinerary-builder>

 - Built a full-stack travel operations platform for multi-city itinerary creation, booking workflows, catalog management, pricing, payments, and document generation.
 - Implemented role-based workflows for super admin, admin, and agent users, including catalog approvals and itinerary versioning.
 - Added generated itinerary, voucher, and invoice PDFs with repeatable local development setup through Docker Compose.